

T0 Projection onto Scramblon Physics: A Quantitative
Bridge between Geometric Spacetime Correction and
Quantum Many-Body Chaos

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Abstract

We present a quantitative comparative analysis between the T0 Time-Mass Duality theory and scramblon physics, motivated by the work of Li, Zhou, Zhang et al. (Phys. Rev. Lett. **136**, 060403, 2026), who achieved the first experimental extraction of the quantum Lyapunov exponent in a many-body system. We show that the T0 damping factor $\mathcal{D}(N) = \exp(-\xi \ln(N)/\Delta f)$ provides a complementary, parameter-free description operating at the same order of magnitude ($\sim 10^{-4}$) as the scramblon corrections. The central testable prediction is a $\ln(N)$ -scaling residuum after complete scramblon error correction, which qualitatively differs from standard error models ($1/N$ or e^{-N}). Consistency is demonstrated across different platforms: NMR spin systems (adamantane, $N = 16$) and superconducting qubit registers ($N = 73$ and 127 , theoretical prediction values). Both approaches – scramblons for dynamics, T0 for geometry – are complementary, not contradictory.

.1 Introduction

Motivation

Li et al. [1] have achieved a milestone in experimental quantum chaos research with their work *Error-resilient Reversal of Quantum Chaotic Dynamics Enabled by Scramblons*: the first extraction of the quantum Lyapunov exponent in a many-body system with macroscopic degrees of freedom. Using solid-state nuclear magnetic resonance (NMR) on adamantane ($C_{10}H_{16}$), they measure the out-of-time-ordered correlator (OTOC) and validate scramblon theory as a universal framework for information scrambling.

This document investigates how the T0 Time-Mass Duality theory [2] relates to these results. We show that the universal T0 correction factor – derived from fractal spacetime geometry with **zero free parameters** – provides a complementary description making consistent predictions at the same order of magnitude.

Core Question

The central question is not whether T0 *replaces* scramblon theory, but whether both frameworks describe different aspects of the same phenomenon:

- **Scramblon theory**: Dynamic description – how fast quantum chaos grows in a system
- **T0 theory**: Geometric description – what fundamental, system-size-dependent damping follows from fractal spacetime structure

Structure

Section .2 describes the experimental system. Section .3 derives the T0 projection. Section .4 compares both frameworks quantitatively. Section .5 demonstrates cross-platform consistency. Section .6 presents T0 predictions for the Lyapunov exponent. Section .7 formulates the testable prediction.

.2 The Experimental System

Adamantane NMR Setup

Li et al. use powdered adamantane ($C_{10}H_{16}$) in a 9.4 T magnetic field. The relevant system parameters are:

Table 1: Experimental parameters of the NMR system (Li et al.)

Parameter	Value	Significance
Sample	Adamantane ($C_{10}H_{16}$)	Plastic crystal
Spins per molecule	16	Protons (1H)
Magnetic field B_0	9.4 T	Superconducting
Larmor frequency	400.2 MHz	1H at 9.4 T
Dipolar coupling	~6.4 kHz	Intermolecular
Char. time τ_c	~156 μs	$1/f_{dipolar}$

Each adamantane molecule contains 16 proton spins. Rapid thermal molecular motion averages out intramolecular interactions; the remaining intermolecular dipolar coupling

(~6.4 kHz) serves as the dominant spin-spin interaction. The sample is placed in a uniform magnetic field that aligns the spin quantization axes.

Scramblon Theory: Summary

Scramblon theory [5, 6, 7] identifies *scramblons* as a novel type of collective excitation in quantum systems with all-to-all connectivity. Their key properties:

- Scramblons mediate the spread of quantum information, analogous to phonons mediating density fluctuations
- The scramblon propagator grows *exponentially* in time, dominating the late-time behavior of the OTOC
- In systems with holographic duality, scramblons in the boundary quantum theory correspond to gravitons in the bulk gravity

The central fitting ansatz of Li et al. takes the form:

$$F_{\alpha\gamma}(\phi, t) = f(e^{\lambda t}, a, b_\alpha, b_\gamma, \phi) \quad (1)$$

where λ is the Lyapunov exponent, a the imperfection coefficient, and b_α, b_γ are operator-dependent coefficients – a total of **4+ free parameters** per Hamiltonian.

.3 T0 Projection onto the NMR Number Space

The Universal Parameter and Its Projection

The fundamental T0 parameter $\xi = \frac{4}{30000} \approx 1.333 \times 10^{-4}$ is universal. How it manifests in a concrete physical system depends on the respective *number space* – the mathematical structure in which it operates.

Definition .3.1 (T0 Damping Factor for N -Spin Systems). For a system with N correlated spins, the T0 damping factor reads:

$$\mathcal{D}(N) = \exp\left(-\frac{\xi \ln(N)}{\Delta f}\right) = N^{-\xi/\Delta f} \quad (2)$$

with $\Delta f = 3 - \xi = 2.999867$ (fractal dimension) and **zero free parameters**.

Remark .3.2. The crucial point: ξ is not a fit quantity. It follows from T0 theory as a geometric parameter of fractal spacetime. The number space of each system – qubit register, spin ensemble, loop order – only determines *how* ξ enters the calculation.

Numerical Results

Table 2: T0 damping factor for various effective cluster sizes in the adamantane system

N (eff. spins)	$\mathcal{D}(N)$	$1 - \mathcal{D}(N)$	Correction (ppm)
16	0.99987678	1.232×10^{-4}	123.2
32	0.99984597	1.540×10^{-4}	154.0
64	0.99981517	1.848×10^{-4}	184.8
100	0.99979534	2.047×10^{-4}	204.7

The T0 correction is at the $\sim 10^{-4}$ level. Li et al. report that their experimental 95% confidence intervals are also at $\sim 10^{-4}$ (error bars are contained within the data markers). The T0 signature thus operates **exactly at the boundary of their measurement precision**.

.4 Quantitative Comparison: Scramblon Dynamics vs. T0 Geometry

Temporal Structure

Scramblon theory describes the OTOC as a time-dependent function with exponential growth:

$$F(t) \approx 1 - \frac{1}{N} \cdot e^{\kappa t} \quad (\text{early times}) \quad (3)$$

where κ is the Lyapunov exponent, bounded by the Maldacena-Shenker-Stanford bound:

$$\kappa \leq \frac{2\pi k_B T}{\hbar} \quad (4)$$

The T0 damping factor $\mathcal{D}(N)$ is, by contrast, *time-independent* – it describes a fundamental geometric correction, not dynamics.

Table 3: Time comparison: scramblon OTOC decay vs. static T0 correction ($N = 16$, $\kappa \cdot \tau_c \approx 1$)

t/τ_c	OTOC scrambling	T0 correction	Ratio T0/Scr.
0.1	6.907×10^{-2}	1.232×10^{-4}	0.0018
0.5	1.030×10^{-1}	1.232×10^{-4}	0.0012
1.0	1.699×10^{-1}	1.232×10^{-4}	0.0007
2.0	4.618×10^{-1}	1.232×10^{-4}	0.0003
3.0	1.000 (saturated)	1.232×10^{-4}	–
5.0	1.000 (saturated)	1.232×10^{-4}	–

Remark .4.1 (Complementarity). Scramblon dynamics and T0 geometry operate on different levels:

- **Scramblons:** Describe *how fast* information spreads (time-dependent, exponential)
- **T0:** Describes *what fundamental limit* exists (time-independent, geometric)

The T0 correction never dominates: already at early times ($t \ll \tau_c$) the scramblon signal $(1/N)e^{\kappa t} \geq 1/16$ is more than two orders of magnitude larger, and at late times it keeps growing exponentially. The ratio T0/scr. stays ≤ 0.0018 for all $t \geq 0$ (Tab. 3). Both effects coexist.

Why Li et al. Do Not Separately Observe the T0 Effect

There are three reasons why the T0 correction is not identified as a separate effect in the data of Li et al.:

1. **Magnitude:** The T0 correction ($\sim 1.2 \times 10^{-4}$) lies exactly at the edge of the experimental confidence intervals ($\sim 10^{-4}$).
2. **Absorption in fits:** The scramblon fitting ansatz (Eq. 1) has 4+ free parameters that can readily absorb a static correction of this magnitude.
3. **No variation:** Li et al. did not systematically vary the effective cluster size N , which leaves the $\ln(N)$ scaling of the T0 correction invisible.

.5 Consistency Across Different Platforms

Same Formula, Different Number Spaces

A central feature of T0 theory: The same formula (Eq. 2), applied to different physical systems (= different number spaces), yields consistent corrections.

Table 4: T0 correction factor across different platforms

System	N	$\mathcal{D}(N)$	$1 - \mathcal{D}(N)$ (ppm)	CHSH (T0)
Adamantane NMR (molecule)	16	0.99987678	123.2	2.828079
Adamantane NMR (cluster)	64	0.99981517	184.8	2.827904
Superconducting (prediction)	73	0.99980932	190.7	2.827888
Superconducting (prediction)	127	0.99978472	215.3	2.827818

Cross-platform consistency of the formula

The transition from NMR spins ($N = 16$) to superconducting qubits ($N = 127$) represents a change of the entire physical system – different hardware, different couplings, different error sources. Yet the same single-line formula describes the theoretical corrections consistently. The CHSH(T0) column contains *theoretical* prediction values, not measurements. Actual hardware measurements (Doc. 147 [3], IBM Heron r2, May 2026) give $S = 2.74$ – the predicted ξ effect ($\sim 10^{-5}$) lies far below the NISQ noise and is not resolvable with present hardware.

.6 T0 Prediction for the Lyapunov Exponent

Modification by Fractal Geometry

Scramblon theory extracts a Lyapunov exponent $\varkappa$ from the experimental data. T0 predicts that this exponent should receive a system-size-dependent correction:

$$\varkappa_{\text{eff}} = \varkappa_{\text{bare}} \cdot N^{-\xi/\Delta f} \quad (5)$$

with the universal exponent $-\xi/\Delta f = -4.445 \times 10^{-5}$.

Table 5: Predicted T0 correction of the Lyapunov exponent

N (eff.)	$\varkappa_{\text{eff}}/\varkappa_{\text{bare}}$	Correction (ppm)
16	0.99987678	123.2
64	0.99981517	184.8
100	0.99979534	204.7
1000	0.99969302	307.0
10000	0.99959072	409.3

Remark .6.1. For typical NMR cluster sizes, the correction is small (~ 100 – 200 ppm), but it grows systematically with $\ln(N)$. For future quantum simulators with $N > 10^4$ effective degrees of freedom, the effect could enter the measurable range.

.7 Testable Prediction

The $\ln(N)$ Residuum

Central Testable Prediction

After complete scramblon error correction (extrapolation to the error-free OTOC), a **residuum** should remain that scales **logarithmically with the cluster size N** :

$$R(N) = 1 - \text{OTOC}_{\text{corrected}}(N) \approx \frac{\xi \cdot \ln(N)}{\Delta f} \quad (6)$$

Numerical values:

$$R(16) \approx 1.23 \times 10^{-4} \quad (7)$$

$$R(100) \approx 2.05 \times 10^{-4} \quad (8)$$

This $\ln(N)$ scaling distinguishes T0 from all standard error models, which typically scale as $1/N$ or e^{-N} .

Experimental Test Protocol

For experimentalists who have access to the data of Li et al. or similar NMR scrambling measurements:

1. **Perform scramblon error correction** (as described in Li et al.) and extract the error-free OTOC for various effective cluster sizes N .
2. **Compute the residuum:** $R(N) = 1 - \text{OTOC}_{\text{corrected}}(N)$
3. **Plot $R(N)$ versus $\ln(N)$.** T0 predicts a linear relationship with slope $\xi/\Delta f \approx 4.44 \times 10^{-5}$.
4. **Cross-platform comparison:** Apply the same analysis on different hardware platforms (superconducting, ion trap, photonic). The residuum should be platform-dependent, but the $\ln(N)$ scaling universal.

The barrier to entry is minimal: The Zenodo raw data from Li et al. are publicly available (DOI: 10.5281/zenodo.16142455), and the T0 correction is a single line of code:

$$\mathcal{D}(N) = \exp\left(-\frac{4/30000 \cdot \ln(N)}{3 - 4/30000}\right) \quad (9)$$

.8 Method Comparison

The only hardware measurement (Doc. 147, $S = 2.74$) shows decoherence; the ξ effect ($\sim 10^{-5}$) cannot be resolved with current hardware.

.9 Conclusion

This analysis demonstrates that T0 theory and the scramblon physics of Li et al. provide **complementary, not contradictory** descriptions of the same phenomenon:

1. **Scramblons** describe the *dynamics*: how fast quantum chaos grows in a system. They require a complete effective field theory apparatus with multiple fit parameters.

Table 6: Systematic comparison: Scramblon theory vs. T0 projection

Property	Scramblon Theory (Li et al.)	T0 Projection
Foundation	Effective field theory	Geometric correction factor
Free parameters	4+ (fits per Hamiltonian)	0 (derived from ξ)
Expression	Multi-step fitting ansatz	$\mathcal{D}(N) = e^{-\xi \ln(N)/\Delta f}$
Describes	Dynamics (chaos rate)	Geometry (fundamental limit)
Time-dependent?	Yes (exponential)	No (static correction)
Platform	NMR-specific	Universal
Validated?	Yes (adamantane)	No (see Doc. 147)

2. **T0** describes the *geometry*: what fundamental, system-size-dependent damping follows from fractal spacetime structure. It requires a single universal parameter and one line of mathematics.
3. Li et al. are **experimentally grounded** in NMR measurements; T0 provides a theoretical correction at the same order of magnitude ($\sim 10^{-4}$). An experimental grounding of T0 in hardware data does not exist: the only hardware measurement (Doc. 147, May 2026) gives $S = 2.74$, and the ξ effect cannot be resolved with present hardware.
4. A **testable prediction** has been identified: the $\ln(N)$ -scaling residuum after scramblon error correction. This prediction can be verified using publicly available raw data.

The analogy to historical parallels is apt: Just as Heisenberg's matrix mechanics and Schrödinger's wave mechanics provided different mathematical descriptions of the same physics, scramblons and T0 may represent different projections of a deeper structure. The experimental distinction lies in the scaling behavior – $\ln(N)$ for T0 vs. $1/N$ for standard error models – and is testable with current technology.

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